



Glossary terms from module 2

Terms and definitions for Course 6, Module 2

Area chart: A data visualization that uses individual data points for a changing variable connected by a continuous line with a filled in area underneath

Box plot: A data visualization that displays the distribution of values along an x-axis

Bubble chart: A data visualization that displays individual data points as bubbles, comparing numeric values by their relative size

Bullet graph: A data visualization that displays data as a horizontal bar chart moving toward a desired value

Circle view: A data visualization that shows comparative strength in data

Column chart: A data visualization that uses individual data points for a changing variable, represented as vertical columns

Combo chart: A data visualization that combines more than one visualization type

Density map: A data visualization that represents concentrations, with color representing the number or frequency of data points in a given area on a map

Distribution graph: A data visualization that displays the frequency of various outcomes in a sample

Diverging color palette: A color theme that displays two ranges of data values using two different hues, with color intensity representing the magnitude of the values

Donut chart: A data visualization where segments of a ring represent data values adding up to a whole

Filled map: A data visualization that colors areas in a map based on measurements or dimensions

Gantt chart: A data visualization that displays the duration of events or activities on a timeline

Gauge chart: A data visualization that shows a single result within a progressive range of values

Highlight table: A data visualization that uses conditional formatting and color on a table

Packed bubble chart: A data visualization that displays data in clustered circles

Symbol map: A data visualization that displays a mark over a given longitude and latitude

Terms and definitions from previous modules

A

A/B testing: The process of testing two variations of the same web page to determine which page is more successful at attracting user traffic and generating revenue

Absolute reference: A reference within a function that is locked so that rows and columns won't change if the function is copied

Access control: Features such as password protection, user permissions, and encryption that are used to protect a spreadsheet

Accuracy: The degree to which data conforms to the actual entity being measured or described

Action-oriented question: A question whose answers lead to change

Administrative metadata: Metadata that indicates the technical source of a digital asset

Agenda: A list of scheduled appointments

Aggregation: The process of collecting or gathering many separate pieces into a whole

Algorithm: A process or set of rules followed for a specific task

Aliasing: Temporarily naming a table or column in a query to make it easier to read and write

Alternative text: Text that provides an alternative to non-text content, such as images and videos

Analytical skills: Qualities and characteristics associated with using facts to solve problems

Analytical thinking: The process of identifying and defining a problem, then solving it by using data in an organized, step-by-step manner

Annotation: Text that briefly explains data or helps focus the audience on a particular aspect of the data in a visualization

Array: A collection of values in spreadsheet cells

Attribute: A characteristic or quality of data used to label a column in a table

Audio file: Digitized audio storage usually in an MP3, AAC, or other compressed format

AVERAGE: A spreadsheet function that returns an average of the values from a selected range

AVERAGEIF: A spreadsheet function that returns the average of all cell values from a given range that meet a specified condition

B

Bad data source: A data source that is not reliable, original, comprehensive, current, and cited (ROCCC)

Balance: The design principle of creating aesthetic appeal and clarity in a data visualization by evenly distributing visual elements

Bar graph: A data visualization that uses size to contrast and compare two or more values

Bias: A conscious or subconscious preference in favor of or against a person, group of people, or thing

Big data: Large, complex datasets typically involving long periods of time, which enable data analysts to address far-reaching business problems

Boolean data: A data type with only two possible values, usually true or false

Borders: Lines that can be added around two or more cells on a spreadsheet

Business task: The question or problem data analysis resolves for a business

C

Calculated field: A new field within a pivot table that carries out certain calculations based on the values of other fields

Calculus: A branch of mathematics that involves the study of rates of change and the changes between values that are related by a function

CASE: A SQL statement that returns records that meet conditions by including an if/then statement in a query

CAST: A SQL function that converts data from one datatype to another

Causation: When an action directly leads to an outcome, such as a cause-effect relationship

Cell reference: A cell or a range of cells in a worksheet typically used in formulas and functions

Changelog: A file containing a chronologically ordered list of modifications made to a project

Channel: A visual aspect or variable that represents characteristics of the data in a visualization

Chart: A graphical representation of data from a worksheet

Clean data: Data that is complete, correct, and relevant to the problem being solved

Cloud: A place to keep data online, rather than a computer hard drive

Cluster: A collection of data points on a data visualization with similar values

COALESCE: A SQL function that returns non-null values in a list

Compatibility: How well two or more datasets are able to work together

Completeness: The degree to which data contains all desired components or measures

CONCAT: A SQL function that adds strings together to create new text strings that can be used as unique keys

CONCATENATE: A spreadsheet function that joins together two or more text strings

Conditional formatting: A spreadsheet tool that changes how cells appear when values meet specific conditions

Confidence interval: A range of values that conveys how likely a statistical estimate reflects the population

Confidence level: The probability that a sample size accurately reflects the greater population

Confirmation bias: The tendency to search for or interpret information in a way that confirms pre-existing beliefs

Consent: The aspect of data ethics that presumes an individual's right to know how and why their personal data will be used before agreeing to provide it

Consistency: The degree to which data is repeatable from different points of entry or collection

Context: The condition in which something exists or happens

Continuous data: Data that is measured and can have almost any numeric value

CONVERT: A SQL function that changes the unit of measurement of a value in data

Cookie: A small file stored on a computer that contains information about its users

Correlation: The measure of the degree to which two variables change in relationship to each other

COUNT: A spreadsheet function that counts the number of cells in a range

COUNTA: A spreadsheet function that counts the total number of values within a range that meet specified criteria

COUNTIF: A spreadsheet function that returns the number of cells in a range that match a specified value

COUNT DISTINCT: A SQL function that only returns the distinct values in a specified range

CREATE TABLE: A SQL clause that adds a temporary table to a database that can be used by multiple people

Cross-field validation: A process that ensures certain conditions for multiple data fields are satisfied

CSV (comma-separated values) file: A delimited text file that uses a comma to separate values

Currency: The aspect of data ethics that presumes individuals should be aware of financial transactions resulting from the use of their personal data and the scale of those transactions

D

Dashboard: A tool that monitors live, incoming data

Data: A collection of facts

Data aggregation: The process of gathering data from multiple sources and combining it into a single, summarized collection

Data analysis: The collection, transformation, and organization of data in order to draw conclusions, make predictions, and drive informed decision-making

Data analysis process: The six phases of ask, prepare, process, analyze, share, and act whose purpose is to gain insights that drive informed decision-making

Data analyst: Someone who collects, transforms, and organizes data in order to draw conclusions, make predictions, and drive informed decision-making

Data analytics: The science of data

Data anonymization: The process of protecting people's private or sensitive data by eliminating identifying information

Data bias: When a preference in favor of or against a person, group of people, or thing systematically skews data analysis results in a certain direction

Data composition: The process of combining the individual parts in a visualization and displaying them together as a whole

Data constraints: The criteria that determine whether a piece of a data is clean and valid

Data design: How information is organized

Data-driven decision-making: Using facts to guide business strategy

Data ecosystem: The various elements that interact with one another in order to produce, manage, store, organize, analyze, and share data

Data element: A piece of information in a dataset

Data engineer: A professional who transforms data into a useful format for analysis and gives it a reliable infrastructure

Data ethics: Well-founded standards of right and wrong that dictate how data is collected, shared, and used

Data governance: A process for ensuring the formal management of a company's data assets

Data-inspired decision-making: Exploring different data sources to find out what they have in common

Data integrity: The accuracy, completeness, consistency, and trustworthiness of data throughout its life cycle

Data interoperability: The ability to integrate data from multiple sources and a key factor leading to the successful use of open data among companies and governments

Data life cycle: The sequence of stages that data experiences, which include plan, capture, manage, analyze, archive, and destroy

Data manipulation: The process of changing data to make it more organized and easier to read

Data mapping: The process of matching fields from one data source to another

Data merging: The process of combining two or more datasets into a single dataset

Data model: A tool for organizing data elements and how they relate to one another

Data privacy: Preserving a data subject's information any time a data transaction occurs

Data range: Numerical values that fall between predefined maximum and minimum values

Data replication: The process of storing data in multiple locations

Data science: A field of study that uses raw data to create new ways of modeling and understanding the unknown

Data security: Protecting data from unauthorized access or corruption by adopting safety measures

Data strategy: The management of the people, processes, and tools used in data analysis

Data transfer: The process of copying data from a storage device to computer memory or from one computer to another

Data type: An attribute that describes a piece of data based on its values, its programming language, or the operations it can perform

Data validation: A tool for checking the accuracy and quality of data

Data validation process: The process of checking and rechecking the quality of data so that it is complete, accurate, secure and consistent

Data visualization: The graphical representation of data

Data warehousing specialist: A professional who develops processes and procedures to effectively store and organize data

Database: A collection of data stored in a computer system

Dataset: A collection of data that can be manipulated or analyzed as one unit

DATEDIF: A spreadsheet function that calculates the number of days, months, or years between two dates

Decision tree: A tool that helps analysts make decisions about critical features of a visualization

Delimiter: A character that indicates the beginning or end of a data item

Descriptive metadata: Metadata that describes a piece of data and can be used to identify it at a later point in time

Design thinking: A process used to solve complex problems in a user-centric way

Digital photo: An electronic or computer-based image usually in BMP or JPG format

Dirty data: Data that is incomplete, incorrect, or irrelevant to the problem to be solved

Discrete data: Data that is counted and has a limited number of values

DISTINCT: A keyword that is added to a SQL SELECT statement to retrieve only non-duplicate entries

Distribution graph: A data visualization that displays the frequency of various outcomes in a sample

DROP TABLE: A SQL clause that removes a temporary table from a database

Duplicate data: Any record that inadvertently shares data with another record

Dynamic visualizations: Data visualizations that are interactive or change over time

E

Emphasis: The design principle of arranging visual elements to focus the audience's attention on important information in a data visualization

Equation: A calculation that involves addition, subtraction, multiplication, or division (also called a math expression)

Estimated response rate: The average number of people who typically complete a survey

Ethics: Well-founded standards of right and wrong that prescribe what humans ought to do, usually in terms of rights, obligations, benefits to society, fairness, or specific virtues

Experimenter bias: The tendency for different people to observe things differently (Refer to Observer bias)

External data: Data that lives, and is generated, outside of an organization

F

Fairness: A quality of data analysis that does not create or reinforce bias

Field: A single piece of information from a row or column of a spreadsheet; in a data table, typically a column in the table

Field length: A tool for determining how many characters can be keyed into a spreadsheet field

Fill handle: A box in the lower-right-hand corner of a selected spreadsheet cell that can be dragged through neighboring cells in order to continue an instruction

Filtering: The process of showing only the data that meets a specified criteria while hiding the rest

Find and replace: A tool that finds a specified search term and replaces it with something else

First-party data: Data collected by an individual or group using their own resources

Float: A number that contains a decimal

Foreign key: A field within a database table that is a primary key in another table (Refer to primary key)

Formula: A set of instructions used to perform a calculation using the data in a spreadsheet

FROM: The section of a query that indicates from which table(s) to extract the data

Function: A preset command that automatically performs a specified process or task using the data in a spreadsheet

G

Gap analysis: A method for examining and evaluating the current state of a process in order to identify opportunities for improvement in the future

General Data Protection Regulation of the European Union (GDPR): Policy-making body in the European Union created to help protect people and their data

Geolocation: The geographical location of a person or device by means of digital information

Good data source: A data source that is reliable, original, comprehensive, current, and cited (ROCCC)

GROUP BY: A SQL clause that groups rows that have the same values from a table into summary rows

H

HAVING: A SQL clause that adds a filter to a query instead of the underlying table that can only be used with aggregate functions

Header: The first row in a spreadsheet that labels the type of data in each column

Headline: Text at the top of a visualization that communicates the data being presented

Heat map: A data visualization that uses color contrast to compare categories in a dataset

Histogram: A data visualization that shows how often data values fall into certain ranges

Hypothesis testing: A process to determine if a survey or experiment has meaningful results

I

Incomplete data: Data that is missing important fields

Inconsistent data: Data that uses different formats to represent the same thing

Incorrect/inaccurate data: Data that is complete but inaccurate

INNER JOIN : A SQL function that returns records with matching values in both tables

Inner query: A SQL subquery that is inside of another SQL statement

Internal data: Data that lives within a company's own systems

Interpretation bias: The tendency to interpret ambiguous situations in a positive or negative way

J

JOIN: A SQL function that is used to combine rows from two or more tables based on a related column

K

L

Label: Text in a visualization that identifies a value or describes a scale

Leading question: A question that steers people toward a certain response

LEFT: A function that returns a set number of characters from the left side of a text string

LEFT JOIN: A SQL function that will return all the records from the left table and only the matching records from the right table

Legend: A tool that identifies the meaning of various elements in a data visualization

LEN: A function that returns the length of a text string by counting the number of characters it contains

Length: The number of characters in a text string

LIMIT: A SQL clause that specifies the maximum number of records returned in a query

Line graph: A data visualization that uses one or more lines to display shifts or changes in data over time

Long data: A dataset in which each row is one time point per subject, so each subject has data in multiple rows

M

Mandatory: A data value that cannot be left blank or empty

Map: A data visualization that organizes data geographically

Margin of error: The maximum amount that sample results are expected to differ from those of the actual population

Mark: A visual object in a data visualization such as a point, line, or shape

MATCH: A spreadsheet function used to locate the position of a specific lookup value

Math expression: A calculation that involves addition, subtraction, multiplication, or division (also called an equation)

Math function: A function that is used as part of a mathematical formula

MAX: A function that returns the largest numeric value from a range of cells

MAXIFS: A spreadsheet function that returns the maximum value from a given range that meets a specified condition

Measurable question: A question whose answers can be quantified and assessed

Mental model: A data analyst's thought process and approach to a problem

Mentor: Someone who shares knowledge, skills, and experience to help another grow both professionally and personally

Merger: An agreement that unites two organizations into a single new one

Metadata: Data about data

Metadata repository: A database created to store metadata

Metric: A single, quantifiable type of data that is used for measurement

Metric goal: A measurable goal set by a company and evaluated using metrics

MID: A function that returns a segment from the middle of a text string

MIN: A spreadsheet function that returns the smallest numeric value from a range of cells

MINIFS: A spreadsheet function that returns the minimum value from a given range that meets a specified condition

Modulo: An operator (%) that returns the remainder when one number is divided by another

Movement: The design principle of arranging visual elements to guide the audience's eyes from one part of a data visualization to another

N

Naming conventions: Consistent guidelines that describe the content, creation date, and version of a file in its name

Narrative: (Refer to story)

Networking: Building relationships by meeting people both in person and online

Nominal data: A type of qualitative data that is categorized without a set order

Normalized database: A database in which only related data is stored in each table

Notebook: An interactive, editable programming environment for creating data reports and showcasing data skills

Null: An indication that a value does not exist in a dataset

O

Observation: The attributes that describe a piece of data contained in a row of a table

Observer bias: The tendency for different people to observe things differently (also called experimenter bias)

Open data: Data that is available to the public

Openness: The aspect of data ethics that promotes the free access, usage, and sharing of data

Operator: A symbol that names the operation or calculation to be performed

ORDER BY: A SQL clause that sorts results returned in a query

Order of operations: Using parentheses to group together spreadsheet values in order to clarify the order in which operations should be performed

Ordinal data: Qualitative data with a set order or scale

Outdated data: Any data that has been superseded by newer and more accurate information

OUTER JOIN: A SQL function that combines RIGHT and LEFT JOIN to return all matching records in both tables

Outer query: A SQL statement containing a subquery

Ownership: The aspect of data ethics that presumes individuals own the raw data they provide and have primary control over its usage, processing, and sharing

P

Pattern: The design principle of using similar visual elements to demonstrate trends and relationships in a data visualization

Pie chart: A data visualization that uses segments of a circle to represent the proportions of each data category compared to the whole

Pivot chart: A chart created from the fields in a pivot table

Pivot table: A data summarization tool used to sort, reorganize, group, count, total, or average data

Pixel: In digital imaging, a small area of illumination on a display screen that, when combined with other adjacent areas, forms a digital image

Population: In data analytics, all possible data values in a dataset

Pre-attentive attributes: The elements of a data visualization that an audience recognizes automatically without conscious effort

Primary key: An identifier in a database that references a column in which each value is unique (Refer to foreign key)

Problem domain: The area of analysis that encompasses every activity affecting or affected by a problem

Problem types: The various problems that data analysts encounter, including categorizing things, discovering connections, finding patterns, identifying themes, making predictions, and spotting something unusual

Profit margin: A percentage that indicates how many cents of profit has been generated for each dollar of sale

Proportion: The design principle of using the relative size and arrangement of visual elements to demonstrate information in a data visualization

Q

Qualitative data: A subjective and explanatory measure of a quality or characteristic

Quantitative data: A specific and objective measure, such as a number, quantity, or range

Query: A request for data or information from a database

Query language: A computer programming language used to communicate with a database

R

R: A programming language used for statistical analysis, visualization, and other data analysis

Random sampling: A way of selecting a sample from a population so that every possible type of the sample has an equal chance of being chosen

Range: A collection of two or more cells in a spreadsheet

Ranking: A system to position values of a dataset within a scale of achievement or status

Record: A collection of related data in a data table, usually synonymous with row

Redundancy: When the same piece of data is stored in two or more places

Reframing: The process of restating a problem or challenge, then redirecting it toward a potential resolution

Regular expression (RegEx): A rule that says the values in a table must match a prescribed pattern

Relational database: A database that contains a series of tables that can be connected to form relationships

Relativity: The process of considering observations in relation or proportion to something else

Relevant question: A question that has significance to the problem to be solved

Remove duplicates: A spreadsheet tool that automatically searches for and eliminates duplicate entries from a spreadsheet

Repetition: The design principle of repeating visual elements to demonstrate meaning in a data visualization

Report: A static collection of data periodically given to stakeholders

Return on investment (ROI): A formula that uses the metrics of investment and profit to evaluate the success of an investment

Revenue: The total amount of income generated by the sale of goods or services

Rhythm: The design principle of creating movement and flow in a data visualization to engage an audience

RIGHT: A function that returns a set number of characters from the right side of a text string

RIGHT JOIN: A SQL function that will return all records from the right table and only the matching records from the left.

Root cause: The reason why a problem occurs

ROUND: A SQL function that returns a number rounded to a certain number of decimal places

S

Sample: In data analytics, a segment of a population that is representative of the entire population

Sampling bias: Overrepresenting or underrepresenting certain members of a population as a result of working with a sample that is not representative of the population as a whole

Scatterplot: A data visualization that represents relationships between different variables with individual data points without a connecting line

Schema: A way of describing how something, such as data, is organized

Scope of work (SOW): An agreed-upon outline of the tasks to be performed during a project

Second-party data: Data collected by a group directly from its audience and then sold

SELECT: The section of a query that indicates from which column(s) to extract the data

SELECT INTO: A SQL clause that copies data from one table into a temporary table without adding the new table to the database

Small data: Small, specific data points typically involving a short period of time, which are useful for making day-to-day decisions

SMART methodology: A tool for determining a question's effectiveness based on whether it is specific, measurable, action-oriented, relevant, and time-bound

Social media: Websites and applications through which users create and share content or participate in social networking

Soft skills: Nontechnical traits and behaviors that relate to how people work

Sort range: A spreadsheet menu function that sorts a specified range and preserves the cells outside the range

Sort sheet: A spreadsheet menu function that sorts all data by the ranking of a specific sorted column and keeps data together across rows

Sorting: The process of arranging data into a meaningful order to make it easier to understand, analyze, and visualize

Specific question: A question that is simple, significant, and focused on a single topic or a few closely related ideas

SPLIT: A spreadsheet function that divides text around a specified character and puts each fragment into a new, separate cell

Sponsor: A professional advocate who is committed to moving forward the career of another

Spreadsheet: A digital worksheet

SQL: (Refer to Structured Query Language)

Stakeholders: People who invest time and resources into a project and are interested in its outcome

Static visualization: A data visualization that does not change over time unless it is edited

Statistical power: The probability that a test of significance will recognize an effect that is present

Statistical significance: The probability that sample results are not due to random chance

Statistics: The study of how to collect, analyze, summarize, and present data

Story: The narrative of a data presentation that makes it meaningful and interesting

String data type: A sequence of characters and punctuation that contains textual information (also called text data type)

Structural metadata: Metadata that indicates how a piece of data is organized and whether it is part of one or more than one data collection

Structured data: Data organized in a certain format such as rows and columns

Structured Query Language: A computer programming language used to communicate with a database

Structured thinking: The process of recognizing the current problem or situation, organizing available information, revealing gaps and opportunities, and identifying options

Subquery: A SQL query that is nested inside a larger query

SUBSTR: A SQL function that extracts a substring from a string variable

Substring: A subset of a text string

Subtitle: Text that supports a headline by adding context and description

SUM: A function that adds the values of a selected range of cells

SUMIF: A spreadsheet function that adds numeric data based on one condition

Summary table: A table used to summarize statistical information about data

SUMPRODUCT: A function that multiplies arrays and returns the sum of those products

Syntax: The predetermined structure of a language that includes all required words, symbols, and punctuation, as well as their proper placement

T

Tableau: A business intelligence and analytics platform that helps people visualize, understand, and make decisions with data

Technical mindset: The ability to break things down into smaller steps or pieces and work with them in an orderly and logical way

Temporary table: A database table that is created and exists temporarily on a database server

Text data type: A sequence of characters and punctuation that contains textual information (also called string data type)

Text string: A group of characters within a cell, most often composed of letters

Third-party data: Data provided from outside sources who didn't collect it directly

Time-bound question: A question that specifies a timeframe to be studied

Transaction transparency: The aspect of data ethics that presumes all data-processing activities and algorithms should be explainable and understood by the individual who provides the data

Transferable skills: Skills and qualities that can transfer from one job or industry to another

TRIM: A function that removes leading, trailing, and repeated spaces in data

Turnover rate: The rate at which employees voluntarily leave a company

Typecasting: Converting data from one type to another

U

Unbiased sampling: When the sample of the population being measured is representative of the population as a whole

Underscores: Lines used to underline words and connect text characters

Unfair question: A question that makes assumptions or is difficult to answer honestly

Unique: A value that can't have a duplicate

United States Census Bureau: An agency in the U.S. Department of Commerce that serves as the nation's leading provider of quality data about its people and economy

Unity: The design principle of using visual elements that complement each other to create aesthetic appeal and clarity in a data visualization

Unstructured data: Data that is not organized in any easily identifiable manner

V

Validity: The degree to which data conforms to constraints when it is input, collected, or created

VALUE: A spreadsheet function that converts a text string that represents a number to a numeric value

Variety: The design principle of using different kinds of visual elements in a data visualization to engage an audience

Verification: A process to confirm that a data-cleaning effort was well executed and the resulting data is accurate and reliable

Video file: A collection of images, audio files, and other data usually encoded in a compressed format such as MP4, MV4, MOV, AVI, or FLV

Visual form: The appearance of a data visualization that gives it structure and aesthetic appeal

Visualization: (Refer to Data visualization)

VLOOKUP: A spreadsheet function that vertically searches for a certain value in a column to return a corresponding piece of information

W

WHERE: The section of a query that specifies criteria that the requested data must meet

Wide data: A dataset in which every data subject has a single row with multiple columns to hold the values of various attributes of the subject

WITH: A SQL clause that creates a temporary table that can be queried multiple times

World Health Organization: An organization whose primary role is to direct and coordinate international health within the United Nations system

X

X-axis: The horizontal line of a graph usually placed at the bottom, which is often used to represent time scales and discrete categories

Y

Y-axis: The vertical line of a graph usually placed to the left, which is often used to represent frequencies and other numerical variables

Z